

EXAM 7 – SPRING 2013

2. (2.75 points)

Given the following information:

<u>Accident Year</u>	<u>Incremental Paid Loss (\$000)</u>			<u>Premium (\$000)</u>
	<u>0-12</u>	<u>12-24</u>	<u>24-36</u>	
2010	1,400	650	280	3,600
2011	1,000	850		3,700
2012	1,500			3,650
Total	3,900	1,500	280	10,950

Assume that there is no further development beyond 36 months.

a. (2 points)

Calculate the individual loss ratio claims reserve R^{ind} for the total of accident years 2010 through 2012.

b. (0.75 point)

Calculate the collective loss ratio claims reserve R^{coll} for the total of accident years 2010 through 2012.

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Question 2 Sample Answer

Solution 1

a) R^{ind} for total (2010-2012) = $\sum q_i / p_i \times C_i$

$$m_k = \frac{\text{sum(loss @ dev age i)}}{\text{sum(Premium)}}$$

$$m_{0-12} = \frac{1400+1000+1500}{3600+3700+3650} = 0.356$$

$$m_{12-24} = \frac{650+850}{3600+3700} = 0.205$$

$$m_{24-36} = \frac{280}{3600} = 0.078$$

$$p_i = \frac{\sum m_i \text{ to date}}{\sum m_i}$$

$$p_{2010} = \frac{0.356+0.205+0.078}{0.356+0.205+0.078} = 1.000$$

$$p_{2011} = \frac{0.356+0.205}{0.356+0.205+0.078} = 0.878$$

$$p_{2012} = \frac{0.356}{0.356+0.205+0.078} = 0.557$$

$$q_i = 1 - p_i$$

$$q_{2010} = 1 - 1 = 0$$

$$q_{2011} = 1 - 0.878 = 0.122$$

$$q_{2012} = 1 - 0.557 = 0.443$$

$$C_i = \sum \text{paid losses to date}$$

$$C_{2010} = 1,400 + 650 + 280 = 2,330$$

$$C_{2011} = 1,000 + 850 = 1,850$$

$$C_{2012} = 1,500$$

$$R^{\text{ind}}_{2010} = 0/1 \times 2,330 = 0$$

$$R^{\text{ind}}_{2011} = 0.122/0.878 \times 1,850 = 256$$

$$R^{\text{ind}}_{2012} = 0.443/0.557 \times 1,500 = 1,193$$

$$R^{\text{ind}}_{\text{Total}} = 0 + 256 + 1,193 = 1,449$$

b) Process variance = $\sigma^2 \sum_i R_i = 10000(2032979)$

$$\text{Process SD} = \sqrt{(P \text{ Variance})} = 142,582.56$$

$$R^{\text{coll}} \text{ for total} = q_i \times U^{\text{BC}}_i = q_i \times m_{\text{total}} \times V_i$$

$$R^{\text{coll}}_1 = (0 \times .639) \times 3,600 = 0$$

$$\begin{aligned}
R^{\text{coll}}_2 &= (.122 \times .639) \times 3,700 = 288 \\
R^{\text{coll}}_3 &= (.443 \times .639) \times 3,650 = 1,034 \\
R^{\text{coll}}_{\text{Total}} &= 0 + 288 + 1034 = 1,322
\end{aligned}$$

Solution 2

a) R^{ind} for total (2010-2012) = $\sum q_i / p_i \times C_i$

$$m_k = \frac{\text{sum(loss @ dev age } i)}{\text{sum(Premium)}}$$

$$m_{0-12} = \frac{1400+1000+1500}{3600+3700+3650} = 0.356$$

$$m_{12-24} = \frac{650+850}{3600+3700} = 0.205$$

$$m_{24-36} = \frac{280}{3600} = 0.078$$

$$p_i = \frac{\sum m_i \text{ to date}}{\sum m_i}$$

$$p_{2010} = \frac{0.356+0.205+0.078}{0.356+0.205+0.078} = 1.000$$

$$p_{2011} = \frac{0.356+0.205}{0.356+0.205+0.078} = 0.878$$

$$p_{2012} = \frac{0.356}{0.356+0.205+0.078} = 0.557$$

$$q_i = 1 - p_i$$

$$q_{2010} = 1 - 1 = 0$$

$$q_{2011} = 1 - 0.878 = 0.122$$

$$q_{2012} = 1 - 0.557 = 0.443$$

$$C_i = \sum \text{paid losses to date}$$

$$C_{2010} = 1,400 + 650 + 280 = 2,330$$

$$C_{2011} = 1,000 + 850 = 1,850$$

$$C_{2012} = 1,500$$

$$R^{\text{ind}}_{2010} = 2,330 / 1.00 - 2,330 = 0$$

$$R^{\text{ind}}_{2011} = 1,850 / .878 - 1,850 = 256$$

$$R^{\text{ind}}_{2012} = 1,500 / .557 - 1,500 = 1,193$$

$$R^{\text{ind}}_{\text{Total}} = 0 + 256 + 1,193 = 1,449$$

b) $U^{\text{BC}}_i = \text{Premium} \times \text{ELR} = V_i \times \sum m_i$

$$U^{\text{BC}}_{2010} = 3,600 \times (0.356+0.205+0.078) = 2,302$$

$$U^{\text{BC}}_{2011} = 3,700 \times (0.356+0.205+0.078) = 2,366$$

$$U^{\text{BC}}_{2012} = 3,650 \times (0.356+0.205+0.078) = 2,334$$

$$\begin{aligned}
R^{\text{coll}}_1 &= 0 \times 2,302 = 0 \\
R^{\text{coll}}_2 &= .122 \times 2,366 = 288 \\
R^{\text{coll}}_3 &= .443 \times 2,334 = 1,034 \\
R^{\text{coll}}_{\text{Total}} &= 0 + 288 + 1,034 = 1,322
\end{aligned}$$

Solution 3

b)

$$\begin{aligned}
R^{\text{coll}}_1 \quad 0 &= (63.9\% - 35.6\% - 20.5\% - 7.8\%) \times 3,600 \\
R^{\text{coll}}_2 \quad 288 &= (63.9\% - 35.6\% - 20.5\%) \times 3,700 \\
R^{\text{coll}}_3 \quad 1,034 &= (63.9\% - 35.6\%) \times 3,650 \\
R^{\text{coll}}_{\text{Total}} \quad 1,322 &
\end{aligned}$$

Examiner Comment

Several candidates used a standard paid loss development method instead of calculating the individual loss ratio claims reserve R^{ind} as requested. Another common mistake was miscalculating the m_k 's using the total earned premium for all accident years. Another common mistake was candidates misidentifying R^{ind} as R^{coll} , and vice versa. Candidates making these mistakes did not receive full credit.